SESSION 3 — AFTERNOON

Financial Statement Analysis II

Cash flows • Working capital • Ratio analysis • DuPont decomposition

Session Roadmap — 2 Hours

This session completes the FSA toolkit: cash flows, the full ratio suite, and the DuPont decomposition that ties everything together.

01 **Cash Flow Statement**
Why CFS matters • 3 sections • direct vs indirect • free cash flow

02 **Acme Corporation (CFA LM4)**
Full indirect method: NI=2,210 → CFO=2,606 • CFI=(538) • CFF=(2,220)

03 **Working Capital & Activity**
DSO, DIO, DPO • Cash Conversion Cycle • Inditex CCC=−86 days (CFA Vol 3 LM4)

04 **Liquidity, Solvency & Profitability**
All 4 ratio categories • Apple 5yr margins (CFA LM12) • coverage ratios

05 **DuPont Analysis (CFA LM12)**
3-factor ROE decomposition • Anson Industries trend • 5-factor model

06 **Practice & Reference**
5 worked problems • complete formula reference • key takeaways

Learning Objectives

Understand the Cash Flow Statement

Explain the three sections of the CFS. Distinguish direct from indirect methods. Compute free cash flow.

Build Indirect Method CFO (CFA LM4)

Reconstruct Acme Corporation's CFO=2,606 from NI=2,210 through all adjustments. Interpret each line.

Calculate Working Capital Ratios

Compute DSO, DIO, DPO. Calculate CCC. Interpret Inditex CCC=−86 days as a supply chain advantage.

Apply All 4 Ratio Categories

Activity, liquidity, solvency and profitability ratios. Identify contradictions between ratios (CFA LM12 example).

Decompose ROE with DuPont (CFA LM12)

Apply 3-factor DuPont: ROE = margin × asset turnover × leverage. Analyse Anson Industries FY2→FY5 improvement.

Integrate Ratios for Judgement

Explain why no single ratio tells the full story. Use integrated analysis (CFA LM12 Example 13/14) to form conclusions.

SECTION 01

Cash Flow Statement (CFA LM4)

Why the Cash Flow Statement Matters

Net income can be manipulated through accounting choices. Cash flow is harder to fake. CFA LM4: analysts use CFS to evaluate earnings quality and sustainability.

Cash from Operations (CFO) is the most critical number — it shows if the business generates real cash

Revenue recognised ≠ Cash received

Accrual: revenue recorded when earned, cash may arrive months later. Rising AR = gap between NI and CFO. Check: is revenue real or early recognition?

Depreciation reduces NI but NOT cash

D&A is a non-cash charge subtracted from NI. Must add back in CFO. Acme: +USD 1,052 added back. Distorts NI vs cash comparison.

Investing cash flows reveal true capex

CFI shows actual spending on PP&E, acquisitions. Compare to D&A: if Capex >> D&A persistently, asset base is growing. If << D&A, underinvestment.

Financing flows show capital allocation

Dividends paid, debt repaid, buybacks. CFF reveals how management returns capital to shareholders and manages balance sheet.

Three Sections of the Cash Flow Statement

Every cash flow belongs to one of three sections. Classification rules differ slightly between IFRS and US GAAP for interest and dividends.

Operating Activities (CFO)

Cash generated from or used by the core business. Includes: cash from customers, payments to suppliers, salaries, taxes.
TWO methods: Direct (gross receipts/payments) or Indirect (NI → adjust). Both arrive at same CFO.
US GAAP: interest paid/received and dividends received = CFO. IFRS: more flexibility (interest paid can be CFF).

Investing Activities (CFI)

Cash flows from LONG-TERM investments. Includes: purchase/sale of PP&E, acquisitions, purchase/sale of investment securities. Always negative for growing companies (spending on capex). Acme: bought equipment USD(1,300) + sold equipment USD762 = CFI USD(538).

Financing Activities (CFF)

Cash flows related to DEBT and EQUITY capital. Includes: proceeds from borrowing, repayment of debt, dividends paid, share buybacks, share issuances.
US GAAP: dividends paid = CFF. IFRS: dividends paid can be CFO or CFF. Acme: debt repayment (500) + buyback (600) + dividends (1,120) = CFF USD(2,220).

Direct vs Indirect Method — Same CFO, Different Presentation

Both methods produce IDENTICAL total CFO. The difference is purely in PRESENTATION. Most companies use indirect; analysts often prefer direct.

Feature	Direct Method	Indirect Method
Starting point	Gross cash receipts from customers	Net income (bottom of income statement)
Approach	Show each major cash receipt/payment category separately	Adjust NI for non-cash items + working capital changes
Presentation	Cash received from customers, paid to suppliers, paid to employees, etc.	NI + D&A − gains + AR change + AP change...
Transparency	Higher: shows actual operating cash flows	Lower: operating cash flows hidden in adjustments
Prevalence	Rare in practice (primarily IFRS companies in some regions)	Almost universal for listed companies
Required by	IFRS allows both; US GAAP also allows both but indirect more common	Both IFRS and US GAAP allow
Reconciliation	Must also provide reconciliation of NI to CFO	Reconciliation IS the statement

CFA LM4 note: 'Companies often disclose only indirect operating cash flow information but understanding how cash flow information is put together enables analysts to approximate a direct statement.'

SECTION 02

Acme Corporation (CFA LM4)

Acme Corporation — CFA LM4: Source Data

CFA Vol 4, LM4 uses Acme Corporation (a fictitious retailer) throughout. These exhibits are the foundation for all indirect method calculations.

Income Statement Item (2018)	Amount	Balance Sheet Change	Amount
Revenue (net)	USD 23,598	Accounts receivable increase	+ USD 55
Cost of goods sold	11,456	Inventory increase	+ USD 707
Gross profit	12,142	Prepaid expenses decrease	+ USD 23
D&A expense	1,052	Accounts payable increase	+ USD 263
Other operating expenses	3,577	Salary payable increase	+ USD 10
EBIT	3,390	Interest payable decrease	− USD 12
Gain on sale of equipment	+ 205	Income tax payable increase	+ USD 5
Interest expense	− 246	Other accrued liabilities increase	+ USD 22
Net income	USD 2,210	Long-term debt repaid	− USD 500
D&A (non-cash)	+ 1,052	Common stock repurchased	− USD 600
		Dividends paid	− USD 1,120

Acme CFA LM4 — Indirect Method: Net Income to CFO = 2,606

Directly from CFA Vol 4, LM4, Exhibit 25. Every adjustment has a logical explanation rooted in accrual vs cash differences.

Indirect Method Line Item	Amount	Reason for Adjustment
Net income	+ 2,210	Starting point — accrual basis
+ Depreciation expense	+ 1,052	Non-cash charge deducted from NI. Add back.
− Gain on sale of equipment	− 205	Non-operating gain included in NI. Remove (classified in CFI).
− Increase in accounts receivable	− 55	Accrual revenue > cash collected. A/R rose.
− Increase in inventory	− 707	Cash paid for inventory not yet expensed in COGS.
+ Decrease in prepaid expenses	+ 23	Previously paid; now expensed. Cash effect already occurred.
+ Increase in accounts payable	+ 263	Bought inventory on credit — cash not yet paid out.
+ Increase in wages payable	+ 10	Wage expense accrued but not yet paid.
− Decrease in interest payable	− 12	Paid more interest than accrued on P&L.
+ Increase in income tax payable	+ 5	Tax accrued but not yet paid.
+ Increase in other accrued liab.	+ 22	Other expenses accrued but not yet paid.
= Net cash from operations	= 2,606	CFO: Acme generates solid operating cash

Acme CFA LM4 — Investing & Financing Sections

CFI and CFF are identical under both direct and indirect methods. They reveal how Acme deploys capital and returns it to stakeholders.

Cash Flow from Investing (CFI)

Cash received from sale of equipment: + USD 762 (book value was 762 minus accumulated depr.)
Cash paid for new equipment: − USD 1,300

Net CFI = USD (538)

Note: the gain on sale (USD205) was stripped from CFO. The total cash proceeds from the sale (USD762 = gain + net book value) appear in CFI.

Capex USD1,300 > D&A USD1,052: asset base growing.

Cash Flow from Financing (CFF)

Repaid long-term debt: − USD 500
Repurchased common stock: − USD 600
Paid dividends: − USD 1,120

Net CFF = USD (2,220)

Net change in cash = CFO + CFI + CFF
= 2,606 − 538 − 2,220 = USD (152)
Beginning cash: 1,163. Ending cash: 1,011 ✓

Free Cash Flow = CFO − Capital Expenditures
FCF = 2,606 − 1,300 = USD 1,306 for Acme.
FCF is the cash available AFTER maintaining/growing asset base. Used for: debt repayment, dividends, buybacks, acquisitions, retained cash.
Analysts prefer FCF to NI for valuation because it reflects actual cash available to equity holders.

Interpreting CFO vs Net Income — Earnings Quality Signals

CFO / Net Income ratio: > 1.0x = high quality earnings. << 1.0x = aggressive accruals or working capital deterioration.

For Acme: CFO/NI = 2,606/2,210 = 1.18x. This is good — CFO exceeds NI, primarily due to non-cash D&A exceeding working capital buildups.

Pattern	Signal	Example Cause
CFO >> NI consistently	High quality earnings. Cash conversion excellent.	Strong billing and collections. Conservative accounting.
CFO ≈ NI	Normal. NI tracks cash reasonably well.	Balanced business with moderate working capital.
CFO < NI (accruals positive)	Caution. Revenue or earnings may be overstated.	Rising DSO, channel stuffing, aggressive rev rec.
CFO < 0 while NI > 0	Significant red flag. Business consuming cash despite profits.	Working capital trap. Growth depleting cash. Unsustainable.
CFI > 0 repeatedly	Selling assets to fund operations. Value destruction.	Distressed company liquidating the asset base.
CFF >> 0 repeatedly	Heavy reliance on external financing. Cash burn funded by debt/equity.	Pre-revenue startup or over-leveraged growth company.

CFA LM4 Example 5 — Computing Cash Received from Customers

CFA Vol 4, LM4, Example 5. The direct method requires converting accrual line items to cash amounts. A systematic technique used widely by analysts.

Cash received from customers = Revenue ± Change in AR (+ deferred revenue adjustments)

CFA LM4 EXAMPLE 5 — Blue Bayou Advertising Company

Blue Bayou reports: Revenue = USD 50 million. Total expenses = USD 35 million. Net income = USD 15 million. Accounts receivable DECREASED by USD 12 million.

Cash received from customers = Revenue + Decrease in AR (positive because collected more than billed)
= USD 50m + USD 12m = USD 62m (CFA answer C)

Interpretation: AR fell by USD12m — old receivables collected in cash. Cash exceeded revenue by USD12m.

General rule:
+ AR decrease → collected more cash than revenue recognised
− AR increase → less cash received than revenue (Acme: USD 23,598 − USD 55 = USD 23,543 collected)

CFA also tests: cash paid to suppliers = COGS + inventory increase − AP increase. Cash paid for taxes = tax expense − tax payable increase.

Free Cash Flow Variants — FCFF vs FCFE

FCFF = CFO + Interest×(1−t) − Capex FCFE = CFO − Capex + Net Borrowings

Different stakeholders care about different free cash flow measures. FCFF is pre-debt; FCFE is the residual for equity holders specifically.

FCFF (Free Cash Flow to the Firm)

Available to ALL capital providers — debt AND equity.
Starting point: CFO. Add back after-tax interest (remove financing effect).
Subtract capex.

Use: DCF valuation of the whole enterprise (EV).
Discount at WACC.

Acme FCFF = CFO + Interest×(1−t) − Capex = 2,606 + 246×0.7 − 1,300 = USD 1,478

FCFE (Free Cash Flow to Equity)

Available only to EQUITY holders after debt obligations.
Starting point: CFO. Subtract capex. Add net new borrowings (or subtract net repayments).

Use: equity valuation (Gordon Growth / DDM).
Discount at cost of equity (Ke).

Acme FCFE = CFO − Capex + (LT debt change) = 2,606 − 1,300 − 500 = USD 806

WACC discounts FCFF to get enterprise value. Subtract net debt to get equity value. Cost of equity discounts FCFE to get equity value directly.

SECTION 03

Working Capital & Activity Ratios

Activity Ratios — Measuring Operational Efficiency

Activity ratios measure how efficiently a company converts assets into revenue and cash. Lower DSO, lower DIO = better. Higher DPO = better.

Ratio	Formula	Measures	Interpretation
Days Sales Outstanding (DSO)	Avg AR / (Revenue/365)	Collection speed	Lower = faster cash collection. High DSO = risk or aggressive revenue recognition.
Days Inventory on Hand (DIO)	Avg Inventory / (COGS/365)	Inventory holding period	Lower = faster inventory turns. High DIO = moving stock or demand weakness.
Days Payable Outstanding (DPO)	Avg AP / (COGS/365)	Payment terms with suppliers	Higher = better. Longer payment time = f financing. But too long = supplier relation risk.
Total Asset Turnover (TAT)	Revenue / Avg Total Assets	Asset efficiency	Higher = more revenue per unit of assets model preferred. Retailers > 2x; utilities <
Receivables Turnover	Revenue / Avg AR	AR cycles per year	Higher = faster. Inverse: 365/Receivables turnover = DSO.
Inventory Turnover	COGS / Avg Inventory	Inventory cycles per year	Higher = faster. Inverse: 365/Inventory tu DIO.

Cash Conversion Cycle (CCC) — The Working Capital Clock

CCC = DSO + DIO − DPO Number of days cash is tied up in the operating cycle

The CCC measures how long from paying suppliers to collecting from customers. NEGATIVE CCC = suppliers are financing your operations (ideal).

Day 0	Pay supplier for raw materials
+ DIO days	Hold inventory — cash tied up
+ DSO days	Sell goods on credit, wait to collect
− DPO days	Delay paying supplier (free credit period)
= CCC days	Net days cash is tied up. Target: as low or negative as possible.

Target: minimise CCC. Every day reduction in CCC = cash freed = reduced working capital financing needs.

Inditex CCC = −86 Days — CFA Vol 3 LM4

Directly from CFA Vol 3, LM4. Inditex (Zara parent) achieves a NEGATIVE cash conversion cycle — one of the most cited supply chain advantages in the CFA curriculum.

INDITEX GROUP (ZARA) — CFA Vol 3, LM4

DSO (Days Sales Outstanding) = 11 days — Inditex collects from customers very quickly (mostly retail, card payments).
DIO (Days Inventory on Hand) = 93 days — Inventory turns every 93 days (fast fashion, frequent new collections).
DPO (Days Payable Outstanding) = 190 days — Inditex pays its suppliers extremely slowly (190 days!).

CCC = DSO + DIO − DPO = 11 + 93 − 190 = −86 days

Meaning: Inditex collects cash from customers 86 days BEFORE it pays its suppliers. The suppliers are financing Inditex's operations. This negative CCC is a significant competitive advantage.

Comparison: typical retailer CCC = +30 to +60 days. Amazon CCC is also negative (around −30 to −50 days) for similar reasons.

Why Inditex achieves negative CCC

Enormous buying power allows delayed payment terms. Owns manufacturing in Spain/Portugal (but outsources strategically). Rapid inventory rotation via fast fashion model.

Investor implication

Negative CCC companies can GROW without increasing working capital. Each incremental sale generates cash immediately. Superior business model quality signal.

Integrated Ratio Analysis — CFA LM12 Example 13: Contradiction

CFA LM12, Example 13. A Canadian manufacturer shows RISING current ratio but FALLING quick ratio. A contradiction that reveals an inventory problem.

Ratio	Year 8	Year 9	Year 10	Trend Signal
Current Ratio	1.6	1.9	2.1	↑ Looks like IMPROVING liquidity
Quick Ratio	1.0	0.9	0.8	↓ Shows DETERIORATING liquidity
Days Inventory on Hand (DOH)	30	45	55	↑ Inventory building up significantly
Days Sales Outstanding (DSO)	30	28	24	↓ AR being collected faster

CFA LM12 Resolution

Since current ratio and quick ratio share the same denominator (current liabilities), the divergence must be driven by assets included in current but NOT in quick — i.e., INVENTORY and prepaid expenses.

DOH rising from 30 to 55 days: the company is building up inventory faster than it can sell it. Inventory is rising, pushing up the current ratio.
Meanwhile, DSO fell (collecting AR faster). Cash from collections was apparently USED TO BUY MORE INVENTORY (not reduce CL).

Conclusion: liquidity is DETERIORATING despite the better-looking current ratio. The company may have an inventory management problem or demand is weakening.

CCC Across Industries — Business Model Determines Working Capital

The cash conversion cycle varies enormously by industry. Understanding the CFA benchmark ranges is essential for cross-sectional analysis.

Industry	Typical DSO	Typical DIO	Typical DPO	Typical CCC	Interpretation
Grocery retail	5–10 days	15–25 days	30–45 days	−5 to +5 days	Near zero. Quick turnover, paying suppliers faster.
Fashion retail (Inditex)	10–15 days	90–100 days	180–200 days	−90 to −70 days	Negative. Suppliers finance operations substantially.
Technology / SaaS	30–60 days	10–30 days	30–60 days	0 to +30 days	Low inventory. Subscription model tightens cycle.
Heavy manufacturing	45–90 days	60–120 days	30–60 days	60 to +150 days	Long production cycles, large WC needs.
Aerospace / Defence	60–120 days	90–180 days	30–60 days	120 to +240 days	Custom products, long programmes. Very cash-hungry.
Banking / Financial svcs	N/A	N/A	N/A	N/A	CCC not applicable. Assets ARE loans; different metrics.

Analyst rule: always compare CCC to industry peers, not absolute benchmarks. A CCC of +60 days is excellent in aerospace but dangerous in grocery.

SECTION 04

Liquidity, Solvency & Profitability Ratios

Liquidity Ratios — Can the Business Pay Its Near-Term Bills?

Liquidity = ability to meet short-term obligations as they come due

Ratio	Formula	"Good" Level	Limitation
Current Ratio	Current Assets / Current Liabilities	1.5 to 3.0x	Includes slow-to-liquidate inventory and p
Quick Ratio (Acid Test)	(Cash + Mkt Sec + AR) / CL	0.8 to 1.5x	Excludes inventory. Preferred for manufacturing/retail.
Cash Ratio	(Cash + Mkt Securities) / CL	0.2 to 1.0x	Most conservative. Ignores AR. Often too
Defensive Interval	(Cash + Mkt Sec + AR) / Daily Op Expenses	30 to 60+ days	Days of operating expenses the company fund from current liquid assets without n revenue.

Interpreting liquidity in context

A current ratio of 3.0x is NOT always better. It may signal IDLE CASH or EXCESS INVENTORY. Tech companies often run low current ratios because their business model generates cash quickly (negative CCC).

Amazon current ratio 2023 ≈ 1.0x. Apple ≈ 1.0x. Both are highly creditworthy. Context matters: sector norms, cash generation quality, debt maturity profile.

Solvency Ratios — Long-Term Debt-Servicing Ability

Solvency = ability to meet long-term financial obligations. Analysts also use COVERAGE ratios to assess ability to service interest and fixed charges.

Ratio	Formula	Measures
Debt-to-Equity	Total Debt / Total Equity	Financial leverage. How much of equity is covered by debt.
Debt-to-Assets	Total Debt / Total Assets	What % of assets funded by creditors (not equity).
Financial Leverage	Total Assets / Total Equity	Equity multiplier. If = 2.0x: half funded by debt.
Interest Coverage (EBIT/Interest)	EBIT / Interest Expense	How many times interest is covered. < 1.5x = concern.
Fixed Charge Coverage	(EBIT + Lease pmts) / (Interest + Lease pmts)	Broader coverage including lease obligations.
CFO to Total Debt	CFO / Total Debt	Cash-based solvency. Preferred to earnings-based in distress.

SAP solvency recap (CFA LM3)

Long-term debt to equity 2017: 5,034/25,540 = 19.7% (down from 24.6% in 2016 — deleveraging).
Debt-to-equity: (1,561+5,034)/25,540 = 25.8% (down from 31.4% — all three solvency ratios improved in 2017).

Profitability Ratios — How Efficiently is the Business Converting Revenue to Returns?

Profitability ratios measure margins (income statement) and returns on capital deployed (linking IS to BS). All are higher = better.

Ratio	Formula	What It Measures
Gross Profit Margin	Gross Profit / Revenue	Production and pricing efficiency. Before overhead.
Operating Profit Margin (EBIT)	EBIT / Revenue	Core business profitability. Before financing costs.
Net Profit Margin	Net Income / Revenue	Bottom line after all costs, interest and tax.
Return on Assets (ROA)	Net Income / Avg Total Assets	Profit generated per unit of total assets deployed.
Return on Equity (ROE)	Net Income / Avg Shareholders' Equity	Return to equity holders. DuPont: margin×TAT×leverage.
Return on Invested Capital (ROIC)	EBIT(1−t) / (Equity + LT Debt)	Return on the total capital invested in the business.
EBITDA Margin	EBITDA / Revenue	Pre-D&A operating profitability. Used in LBO/credit analysis.

Apple 5-Year Profitability — CFA LM12 Example 12

Directly from CFA Vol 4, LM12, Example 12. Apple's full five-year trend of profitability ratios. Common-size IS approach applied longitudinally.

Margin	2013	2014	2015	2016	2017	CFA Trend Observation
Gross Profit Margin	37.62%	38.59%	40.06%	39.08%	38.47%	Rose to 2015 peak, then eased. Input cost and mix effects.
Operating Margin	28.67%	28.72%	30.48%	27.84%	26.76%	Steady to 2015 then declining. Cost growth outpaced revenue.
Pre-Tax Margin	29.35%	29.26%	31.03%	28.46%	27.96%	Closely tracks operating margin. Minimal other income.

CFA LM12 Analysis

Sales declined in 2016 (iPhone cycle weakness + foreign currency headwinds) and recovered modestly in 2017. Despite the recovery, ALL margins declined in 2017, indicating costs rose faster than sales.

Bottom line: net profit margin was REMARKABLY STABLE around 21–22% across the full five-year period — a testament to Apple's pricing power and cost discipline.

CFA conclusion: the five-year trend shows the margin variability and its causes — a key analyst skill

EBITDA — The Credit Analyst's Core Metric

EBITDA = EBIT + D&A = Operating income before depreciation

EBITDA is widely used in leveraged finance, LBO analysis and credit work as a proxy for operating cash generation. It removes capital structure and accounting choices.

Metric	Formula	Used For	Limitation
EBITDA	EBIT + Depreciation + Amortisation	Credit analysis, LBO, EV multiples	Ignores capex. High D&A industries: EBITDA overstates cash.
EBITDA Margin	EBITDA / Revenue	Comparing operating efficiency across capital structures	Same problem: capex excluded. High D&A firms look better.
Net Debt / EBITDA	(LT Debt + ST Debt − Cash) / EBITDA	Leverage: years of EBITDA to repay debt. < 2x = low; > 5x = high	EBITDA ≠ cash. Cycle-top EBITDA overestimates capacity.

EBITDA in Leveraged Finance

A company with EBITDA of GBP100m and net debt of GBP400m has leverage of 4.0x. Most LBO financings target 4–6x EBITDA leverage at close.

Bond covenants often require Net Debt/EBITDA < 5.5x or EBITDA/Interest > 2.0x. Breach triggers default or restructuring.

CFA note: Warren Buffett has called EBITDA 'earnings before bad stuff' — capex IS a real cost in capital-intensive businesses.

SECTION 05

DuPont Analysis (CFA LM12)

DuPont Analysis — Decomposing ROE into 3 Drivers

ROE = Net Profit Margin × Total Asset Turnover × Financial Leverage

Developed at DuPont in the 1920s. Now the standard tool for understanding WHY a company's ROE is what it is — and how to improve it.

Net Profit Margin = NI / Revenue — Controls

Profitability: how much profit from each dollar of revenue. Driven by pricing power, cost efficiency, tax management.

Total Asset Turnover = Revenue / Avg Assets — Efficiency

Asset efficiency: how much revenue per dollar of assets. High in retail/distribution, low in capital-intensive industries.

Financial Leverage = Avg Assets / Avg Equity — Leverage

Capital structure: how much assets are funded by equity vs debt. Amplifies ROE but also amplifies risk.

Anson Industries DuPont Trend — CFA LM12, Exhibit 32

Directly from CFA Vol 4, LM12. Anson improved ROE from −0.62% (FY2) to +5.92% (FY5). DuPont reveals the DRIVER of that recovery.

Fiscal Year	ROE	= ROA	x Leverage	Net Margin	x Asset Turnover	x Leverage
FY5	+5.92%	3.70%	1.60x	3.33%	1.11x	1.60x
FY4	+1.66%	1.05%	1.58x	1.11%	0.95x	1.58x
FY3	+1.62%	1.05%	1.54x	1.13%	0.93x	1.54x
FY2	−0.62%	−0.39%	1.60x	−0.47%	0.84x	1.60x

CFA LM12 Insight

Leverage was STABLE throughout (around 1.56–1.60x). Therefore, the improvement in ROE from −0.62% to +5.92% was driven ENTIRELY by improvement in ROA.
ROA improved because: Net Profit Margin recovered from −0.47% to 3.33% AND Total Asset Turnover improved from 0.84x to 1.11x.
Both profitability (margin) and efficiency (asset use) improved simultaneously. Leverage was NOT used to inflate ROE.
CFA conclusion: this is a HIGH QUALITY ROE recovery — driven by fundamental business improvement, not financial engineering.

5-Factor DuPont — Full Decomposition

$$ROE = (NI/EBT) \times (EBT/EBIT) \times (EBIT/Revenue) \times (Revenue/Assets) \times (Assets/Equity)$$

The 5-factor model adds tax efficiency and interest burden to the 3-factor. Separates operating profitability from financing and tax effects.

Factor	Ratio	Measures	Typical Range
Tax burden	NI / EBT	(1 − effective tax rate). Higher = tax efficient.	0.60 to 0.80 (lower tax = higher factor)
Interest burden	EBT / EBIT	Fraction of EBIT left after interest. 1.0 = no debt.	0.70 to 0.99 (lower = more interest paid)
EBIT margin	EBIT / Revenue	Operating profitability. Same as operating margin.	Varies widely: 5–35% typical
Asset turnover	Revenue / Avg Assets	Revenue efficiency. Same as 3-factor TAT.	0.3x (utilities) to 3x+ (retail)
Financial leverage	Avg Assets / Avg Equity	Debt amplification. Same as 3-factor leverage.	1.2x (conservative) to 5x+ (banks)

3-factor vs 5-factor

3-factor groups tax and interest into net margin. 5-factor separates them.

Use 5-factor when: comparing companies with very different tax regimes or capital structures, or investigating WHY margins changed.

ROE vs ROA — The Role of Financial Leverage

Two companies with identical ROA can have very different ROE. Leverage amplifies both returns AND risks. Understanding this is central to banking and credit analysis.

ROE = ROA × (Assets / Equity) If ROA > cost of debt, leverage INCREASES ROE

Scenario	ROA	Leverage	ROE	Risk Assessment
Conservative	10%	1.5x	15%	Low financial risk. Buffer against downturns.
Moderate leverage	10%	2.5x	25%	Moderate. ROE boosted but requires consistent ROA.
High leverage	10%	5.0x	50%	High risk. Any ROA decline rapidly erodes equity value.
Leverage hurts	4%	5.0x	20%	If ROA falls to 4%: ROE = 20% on paper but interest coverage may be < 1x.
Leverage destroys	−2%	5.0x	−10%	Negative ROA amplified to larger negative ROE. Insolvency risk.

Banks routinely operate at leverage ratios of 10–12x (ROA ≈ 1–2%, ROE ≈ 10–20%). One reason banks need strict regulation and capital requirements.

Comparing Companies with DuPont — Strategic Archetypes

Different business models achieve the same ROE through completely different combinations of margin, turnover and leverage. DuPont reveals the strategic model.

Business Type	Net Margin	Asset Turnover	Leverage	ROE	Example
Luxury brand / Pharma	25–40%	0.5–1.0x	1.5–2.0x	19–80%	LVMH, Roche. High r low volume.
Grocery / Mass retail	1–3%	2.5–4.0x	2.5–3.5x	7–42%	Walmart, Tesco. Thi margins, massive vo
SaaS / Technology	20–35%	0.7–1.2x	1.2–1.8x	17–75%	Microsoft, Salesforc Margins expand as s grows.
Utility / Infrastructure	8–15%	0.3–0.5x	3.0–5.0x	7–38%	National Grid. Regul assets, high leverage permitted.
Investment bank	5–10%	0.03–0.05x	10–15x	5–75%	JPMorgan. Ultra-thi on huge asset base. leverage.

Key insight for analysts

When comparing two companies' ROE, always decompose. A retailer's ROE may come from extraordinary asset turns; a pharma's from extraordinary margins. Both may be 15% ROE but for very different structural reasons. DuPont tells you WHICH driver to stress-test.

PRACTICE PROBLEMS

Work through each before checking answers.

Q1

Acme Corporation (CFA LM4): Net income=2,210. D&A=+1,052. Gain on equip=205. AR increase=55. Inventory increase=707. Prepaid decrease=23. AP increase=263. Wages payable increase=10. Interest payable decrease=12. Tax payable increase=5. Other accruals=22. Calculate CFO.

Q2

CFA LM4 Example 5 (Blue Bayou): Revenue=USD50m. AR decreased by USD12m. How much cash was received from customers? Explain the logic.

Q3

Inditex (CFA Vol 3 LM4): DSO=11 days, DIO=93 days, DPO=190 days. Calculate the CCC. What does the result mean for Inditex's financing needs?

Q4

Anson Industries DuPont (CFA LM12): FY5 net margin=3.33%, asset turnover=1.11x, leverage=1.60x. (a) Calculate ROA. (b) Calculate ROE. (c) What drove the improvement from FY2 (ROE=−0.62%)?

Q5

A company has: Net income=USD800k, Revenue=USD10m, Total Assets=USD8m (avg), Total Equity=USD3.2m (avg). Calculate the 3-factor DuPont and verify ROE.

Practice — Worked Solutions

A1

CFO = 2,210 + 1,052 − 205 − 55 − 707 + 23 + 263 + 10 − 12 + 5 + 22 = 2,210+1,052=3,262 −205=3,057 −55=3,002 −707=2,295 +23=2,318 +263=2,581 +10=2,591 −12=2,579 +5=2,584 +22=2,606. CFO = USD 2,606. CFA answer confirmed.

A2

Cash received from customers = Revenue + Decrease in AR = 50 + 12 = USD 62m (CFA answer C). Logic: AR fell by 12m because old credit sales were collected in cash. The company collected MORE cash than it recognised as revenue. General rule: decrease in AR is a positive adjustment to revenue to get cash.

A3

CCC = DSO + DIO − DPO = 11 + 93 − 190 = −86 days. Meaning: Inditex collects cash from customers 86 days BEFORE it pays its suppliers. The suppliers finance operations. Each incremental USD of revenue generates cash without requiring additional working capital financing. This is a powerful competitive advantage — allows growth to be self-funding.

A4

(a) ROA = Net margin × Asset turnover = 3.33% × 1.11 = 3.70% (CFA Exhibit 32 confirmed). (b) ROE = ROA × Leverage = 3.70% × 1.60 = 5.92% (CFA confirmed). (c) Leverage was stable at ~1.60x throughout. Improvement driven by: margin recovery from −0.47% to 3.33% AND efficiency improvement from 0.84x to 1.11x asset turnover. High quality recovery.

A5

Net margin = 800k/10m = 8.0%. Asset turnover = 10m/8m = 1.25x. Leverage = 8m/3.2m = 2.5x. ROE = 8.0% × 1.25 × 2.5 = 25.0%. Verify: NI/Equity = 800k/3.2m = 25.0%. ✓ Also: ROA = NI/Assets = 800k/8m = 10.0%. ROE = ROA × Leverage = 10% × 2.5 = 25.0%.

Formula Reference Card — Session 6

Complete ratio and cash flow formulas from both FSA sessions.

Free Cash Flow	FCF = CFO − Capital Expenditures
Cash received from customers	Revenue + Decrease in AR (or minus AR increase)
Cash paid to suppliers	COGS + Inventory increase − AP increase
Days Sales Outstanding (DSO)	Average AR / (Revenue / 365)
Days Inventory on Hand (DIO)	Average Inventory / (COGS / 365)
Days Payable Outstanding (DPO)	Average AP / (COGS / 365)
Cash Conversion Cycle	CCC = DSO + DIO − DPO
Current Ratio	Current Assets / Current Liabilities
Quick Ratio	(Cash + Marketable Securities + AR) / CL
Interest Coverage	EBIT / Interest Expense
Debt-to-Equity	Total Debt / Total Equity
ROA	Net Income / Average Total Assets
ROE	Net Income / Average Shareholders' Equity
DuPont (3-factor)	ROE = Net Margin × Asset Turnover × Financial Leverage
DuPont (5-factor)	ROE = Tax burden × Interest burden × EBIT margin × TAT × Leverage

KEY TAKEAWAYS

Session 6 — Financial Statement Analysis II

- Acme Corporation (CFA LM4): NI=2,210 → CFO=2,606. Add D&A+1,052, strip gain−205, subtract ΔWC assets, add ΔWC liabilities.
- Acme CFI=(538): sold equip+762, bought equip(1,300). CFF=(2,220): debt(500)+buyback(600)+dividends(1,120). Net cash change=(152).
- Blue Bayou (CFA LM4): Cash from customers = Revenue + ΔAR = 50+12 = USD62m. AR decrease = more cash collected than billed.
- Inditex (CFA Vol 3 LM4): CCC = 11+93−190 = −86 days. Suppliers finance operations. Negative CCC = self-funding growth.
- Anson (CFA LM12): ROE = 3.33% × 1.11 × 1.60 = 5.92%. Leverage stable. Improvement from both margin recovery and asset efficiency.
- Apple 5yr (CFA LM12): net margin stable 21–22%. 2015 margins peak; 2016–2017 decline despite revenue recovery.

- DuPont separates 3 ROE drivers: margin (profitability), asset
- DOH rising + quick ratio falling = inventory problem (CFA